

Neural Networks 2

Overview

23 Feb 2026

Terms

- **logits**
 - the quantile function **associated** with the **standard logistic** distribution
- **quantile function**
 - a probability **distribution's** quantile function is the **inverse** of its **cumulative distribution** function
- **softmax**
 - converts a **tuple** of K **real** numbers into a **probability distribution** over K **possible** outcomes
- **attention**
 - a method that **determines** the **importance** of each **component** in a sequence **relative** to the **other components** in that sequence
- **spatio-temporal data**
 - **data** that shows both **spatial** and **temporal** patterning
- **convolutional neural networks**
 - a type of **feedforward** neural network that **learns features** via filter (or **kernel**) **optimization**
- **multi-axes tensors**
 - a **multi-linear tensor** is an algebraic **object** that **describes** a multilinear **relationship** between sets of algebraic **objects** associated with a **vector space**
- **flat vectors**
 - a **vector** bundle that is endowed with a **linear connection** with **vanishing curvature** (i.e. a **flat** connection)
- **recurrent neural networks**
 - **RNNs** are designed for **processing sequential** data, such as **text**, **speech**, and **time series** where the **order** of elements is **important**

<https://en.wikipedia.org/>

Readings

A. Convolutional Networks (pp. 326 through Section 9.3)

- <https://www.deeplearningbook.org/contents/convnets.html>

B. Sequence Modeling: Recurrent and Recursive Nets (Section 2.5 pp. 120-127)

- <https://www.deeplearningbook.org/contents/rnn.html>

Convolutional Networks

Convolutional Networks (CNNs)

- defined as a type of **neural network** with a **grid** structure, like **images** or **time series**
- inspired by **biological** systems
- foundational for **image** and **pattern recognition** tasks

Convolutional Operation

- a mathematical way to **combine** input **data** with a small **set** of learned **weights** (a **kernel** or **filter**) to extract **features**
- the **kernel slides** across the **input grid** and produces **feature maps**

Key Properties of Convolution

- **sparse** interactions
 - **depends** on a small **local region** of the input
 - **limits** the number of **parameters** dramatically **compared** to **fully** connected layers
- **parameter sharing**
 - the **same** kernel **weights** are **reused** across **different** input positions **reducing** the number of **parameters**
 - **enables learning** patterns that **repeat** across the **data**
- **equivalence to translation**
 - handles **patterns** that **shift** in space
 - if the **input shifts**, the **output shifts** in the **same** way
 - helps **detect patterns** anywhere in the **input**

Pooling	• an operation applied after convolution to reduce the spatial size of feature maps • makes the network less sensitive to small shifts in the input and reduces computation • the most common form is max pooling ○ picks the largest value from a small region
Sequence Modeling: Recurrent and Recursive Nets	
Sequence Modeling	• a sequence is a list of vectors $x^{(1)}, x^{(2)}, x^{(3)}, \dots, x^{(\tau)}$, where each element comes one after the other in time or order • sequence modeling is about processing data where order matters (like text, audio, time series)
Recurrent Neural Networks (RNNs)	• process elements one at a time • at each step the network updates: ○ a hidden state $h^{(t)}$, which is its memory. ○ the hidden state combines the current input $x^{(t)}$ and the previous state h^{t-1} • shared processing lets the network handle sequences of variable length
Why RNNs	• traditional neural networks (like CNNs or fully connected nets) process fixed-length inputs • many tasks require understanding order and history The diagram illustrates the internal structure of an RNN cell at time step t. It shows a central box containing the equations $h(t) = f(h(t-1), x(t))$ and $o(t) = g(h(t))$. An arrow labeled 'new state' points to the hidden state $h(t)$, and an arrow labeled 'output' points from the output $o(t)$. A legend box to the right defines the variables: $h(t)$ is the current memory / hidden state, $x(t)$ is the current input, $o(t)$ is the output at time t, and f, g are learned functions.
Unfolding the RNN	• to understand training and computation, we “unfold” the RNN through time ○ view the network at each step as a separate layer connected to the next • makes training easier to reason about

Parameter Sharing in RNNs	• same weights are used at every time step • reduces the number of parameters • allows the network to generalize sequences of different length
Types of RNN Processing	• outputs at each time step ○ predicting the word in a sequence ○ predicts something every step as it processes input • reads entire sequence then outputs once ○ classifying a sentence after ingesting it fully
Teacher Forcing	• in training, RNNs sometimes use the true output from the training data as input for the next step • this helps learning but it is different from how the network runs in practice
RNNs as Probabilistic Models	• RNNs can be viewed as models that define a probability distribution over sequences • model how likely a next element $y^{(t)}$ is given previous inputs
Conditional RNNs	• RNNs can use another sequence as context to predict a target sequence • can generate an output sequence of the same length as the input sequence • ex: in image captioning, an image is input, and the RNN generates words about that image

Async Materials	
How to Speak the Neural Network Lingo	
Strategies of Defining Neural Networks	• satisfy constraints with differential activations • make differential versions of common or useful operations (i.e. lookups) so networks can make use of them in a learnable fashion
Constraints	• examples • all features must be between a particular range • all features must be non-negative • constrained features must be differentiable ○ if constraint function is not differentiable, the chain rule cannot be applied to compute gradients
Activation Base	• a non-linear mathematical operation is applied to the output of each node • the fundamental driver for learning complex patterns • core components ○ introduction of non-linearity into the system ○ a firing decision determines if a neuronal output is relevant ○ gradient flow ensures that gradients remain large during backpropagation, allowing the model to train effectively
Common Activation Functions	• SoftMax ○ a mathematical function used in artificial intelligence to transform logits ○ each value is between 0 and 1 ○ the sum of all values equals exactly 1 • one-hot vectors ○ represents categorical data as binary vectors where only one element is hot, and the rest are cold • negative log likelihood ○ a loss function used to train classification models by minimizing the negative logarithm of the predicted probability

Additional Activation Functions

- rectified linear unit (ReLU)
 - returns **input** if positive, **0** if not
- sigmoid/logistic
 - input values to a range between 0 and 1
 - frequently used in the output layer for binary classification
- hyperbolic tangent (tanh)
 - maps inputs to a range between -1 and 1
- Leaky ReLU

a variant of ReLU that allows a small negative value, helping to prevent "dying neurons"

SoftMax for Attention

- computes a probability distribution from raw, unnormalized attention scores ensuring they sum to 1
- the resultant normalization allows the model to focus on relevant tokens (high values) and ignore other (low values)

$$\sum_{j=1}^m \frac{\exp \text{sim}(\overrightarrow{\phi(x)}, \vec{k}^{(j)})}{\sum_{c=1}^m \exp \text{sim}(\overrightarrow{\phi(x)}, \vec{k}^{(j)})} \vec{v}^{(j)}$$

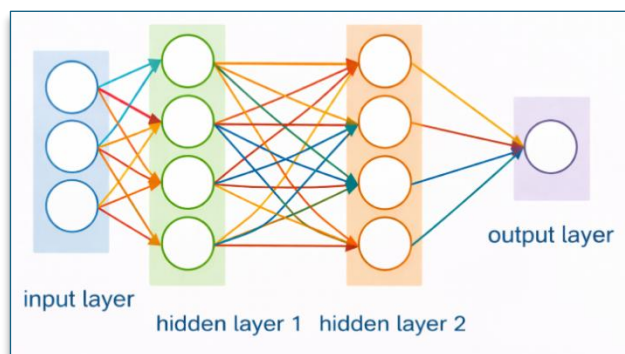
value vectors

keys

Convolutional Neural Networks

Properties

- low-level features are local
- features are translational invariant
- high-level features are composed of low-level features
- modeled after biological analog



ChatGPT 5.2

typical CNN

